NAME ☞	
SCHOOL ☞	
TEACHER ☞	

Pre-Junior Certificate Examination, 2019

Mathematics

Paper 1

Ordinary Level

Time: 2 hours

300 marks

For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5		15	
6			
7			
8			
9			
10		Total	

School Stamp

Running Total	

Grade

Instructions

There are 15 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:	

Question 1 **(Suggested maximum time: 5 minutes)**

Katie wants to buy tickets online for a show.

An adult ticket costs €20
A child ticket costs €15

Katie purchases 6 adult tickets and 4 child tickets.

(a) Work out the **total cost** of the tickets.

(b) VIP seats are available for an additional charge of 15%. Katie selects VIP seats for everyone. Work out the total cost of the tickets when the VIP seat charge is added.

(c) A booking fee of €4·14 was charged for this online service when she paid for the tickets. Write this fee as a **percentage** of the total cost of the tickets including the VIP seat charge.

Question 2 **(Suggested maximum time: 10 minutes)**

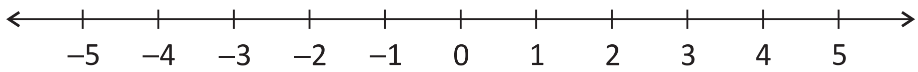
$\boxed{>}$ $\boxed{=}$ $\boxed{<}$

(a) Using the symbols listed above, write the correct symbol in each box to make each of the following statements true. The first one is done for you.

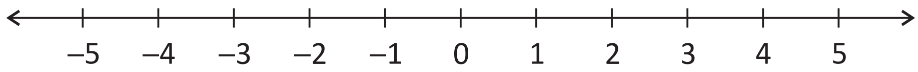
$90 \div 2$	$\boxed{=}$	3×15
18×3	$\boxed{}$	9×6
$64 \div 16$	$\boxed{}$	$17 - 3$
$14 - 27 \div 3$	$\boxed{}$	$\sqrt{5}$
$44 \div 4 + 8$	$\boxed{}$	$14 - 9 \times 5$

(b) Graph each of the following inequalities on the number lines given.
Note: x is an element of a different set ($\mathbb{N}$ or $\mathbb{Z}$) in each case.

(i) $x \leq 3$, where $x \in \mathbb{Z}$.



(ii) $x \leq 3$, where $x \in \mathbb{N}$.



(c) Solve the inequality

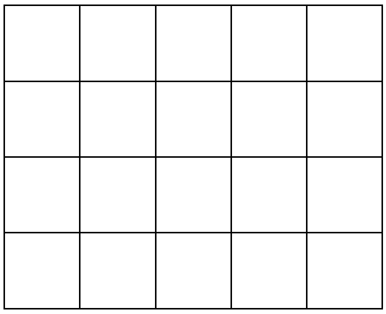
$$3x - 4 < 8, x \in \mathbb{N}$$

and, hence, write down all the values of x which satisfy the inequality.

$x = \boxed{\{ \quad , \quad , \quad \}}$

Question 3 **(Suggested maximum time: 5 minutes)**

(a) Shade in $0{\cdot}7$ of the grid below.



(b) **(i)** Write $\frac{3}{4}$ as a decimal.

(ii) Write $\sqrt{0{\cdot}7}$, correct to two decimal places.

(c) Write $\sqrt{0{\cdot}7}$, $\frac{3}{4}$ and $0{\cdot}7$ in order, from the smallest to the biggest.

Answer = $\square$, $\square$, $\square$.

(d) Write **four hundred thousand** in figures.

(e) A book has 288 pages. Jacob has read $\frac{1}{3}$ of the book.

How many pages are **left** for Jacob to read?

Question 4 **(Suggested maximum time: 5 minutes)**

Karl plans to travel to the Rugby World Cup later this year in Japan.

The first match he plans to attend is Ireland v Scotland on Sunday, 22nd September. The match is due to start at $16{:}45$ local time. The time in Japan is 8 hours ahead of Ireland.

(a) What time will it be in Ireland when the match starts?

(b) Karl plans to purchase a Category B ticket for the match.
The cost of this ticket is $22\,000$ Japanese yen (JPY).
The currency conversion rate is $\mathbf{1}\ \text{JPY} = \mathbf{€0{\cdot}0078}$.
Convert $22\,000$ JPY to euro (€).

(c) The ratio of the cost of a Category B ticket to the cost of a Category D ticket for this match is $\mathbf{11{:}7}$. If Karl purchased a Category D ticket, how much would he save?
Give your answer in euro, correct to the nearest cent.

Question 5 **(Suggested maximum time: 10 minutes)**

(a) Find the value of $(5n + 3) - (2n - 5)$, when $n = 4$.

(b) Patrick says that $x = 5$ is a solution of $x^2 - 9x + 20 = 0$.
Eve says that $x = 4$ is a solution of $x^2 - 9x + 20 = 0$.
Show that they are **both** correct.

(c) Solve the equation $2(2x + 1) = 3(x + 2)$.

Question 6 **(Suggested maximum time: 10 minutes)**

(a) The numbers in the pattern below increase by the same amount each time.
Fill in the boxes to show the missing numbers.

,	42 ,	48 ,	,	60 ,	.

What kind of pattern is made by these numbers?
Tick (✓) **one** box only. **Justify** your answer.

linear ☐ quadratic ☐ exponential ☐

Justification:

(b) The numbers in the pattern below triple each time.
Fill in the boxes to show the missing numbers.

3 ,	9 ,	,	81 ,	.

What kind of pattern is made by these numbers?
Tick (✓) **one** box only. **Justify** your answer.

linear ☐ quadratic ☐ exponential ☐

Justification:

(c) The numbers below form a pattern.

5 ,	15 ,	45 ,	95 ,	165 .

What kind of pattern is made by these numbers?
Tick (✓) **one** box only. **Justify** your answer.

linear ☐ quadratic ☐ exponential ☐

Justification:

Question 7 **(Suggested maximum time: 10 minutes)**

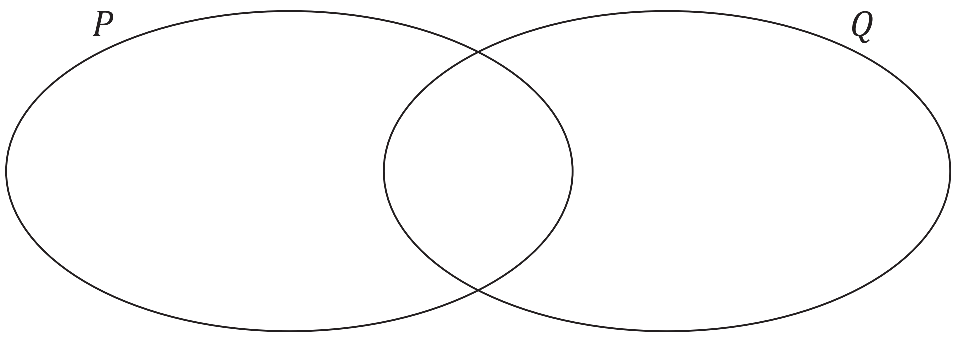
$P = \{m, a, t, h, s\}$
$Q = \{e, x, a, m\}$

(a) Write down **two** different **subsets** of P that have 3 elements each.

Subset 1 = $\{ \quad \}$

Subset 2 = $\{ \quad \}$

(b) Fill in the Venn diagram below to show the sets P and Q.



(c) List the elements of each of the following sets.
Note: Q' is the complement of the set Q.

Set	Elements
$P \cap Q =$	$\{ \quad \}$
$Q \setminus P =$	$\{ \quad \}$
$P \cup Q =$	$\{ \quad \}$
$Q' =$	$\{ \quad \}$

(d) Abby says $\#(P \cap Q) = \#(Q \setminus P)$. Is she correct?
Justify your answer.

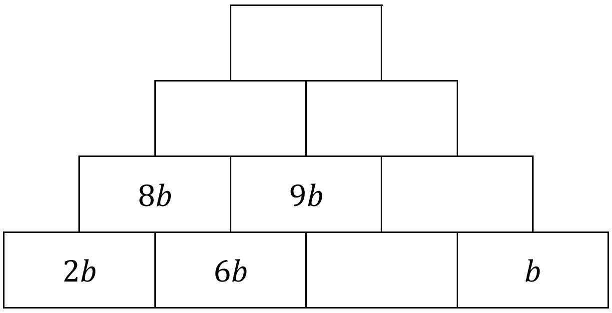
Answer:

Justification:

Question 8 **(Suggested maximum time: 5 minutes)**

(a) In the algebra pyramid below, the two expressions in the lower block are added to find the expression in the block above (for example, $2b + 6b = 8b$).

Complete the algebra pyramid by filling in the empty blocks.



(b) Factorise each of the following:

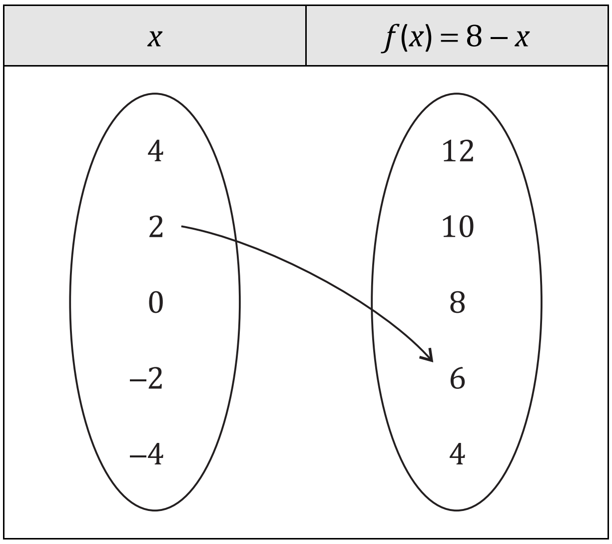
(i) $40v + 16w$

(ii) $ax + ay - bx - by$.

(c) Multiply out and simplify $(3k + 5)(k - 3)$.

Question 9 **(Suggested maximum time: 5 minutes)**

(a) Complete the following mapping diagram of the function $f : x \mapsto 8 - x$.



(b) **(i)** Write down the **domain** of the function $f(x)$, as shown in the diagram.

Domain = $\{ \quad , \quad , \quad , \quad , \quad \}$

(ii) Write down the **range** of the function $f(x)$, as shown in the diagram.

Range = $\{ \quad , \quad , \quad , \quad , \quad \}$

(c) **(i)** Find the value of $f(3)$.

$f(3) =$

(ii) Find the value of $f(-3)$.

$f(-3) =$

Question 10 **(Suggested maximum time: 5 minutes)**

Solve the following simultaneous equations.

$$\begin{aligned} 3x + 2y &= 12 \\ 4x - y &= 5 \end{aligned}$$

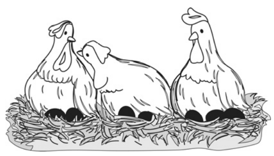
Question 11 **(Suggested maximum time: 5 minutes)**

(a) The table below shows the approximate areas of Ireland and Germany.

Country	**Area (square kilometres)**
Ireland	$71\,000$
Germany	$355\,000$

The area of Germany is larger than the area of Ireland.
How many times larger is Germany?

(b) In January, Dara collected 3, 4 or 5 eggs each day, from his hens.
In the first 15 days he collected 64 eggs altogether.



Write down the **least** number of eggs that Dara collected in January and the **greatest** number of eggs that Dara collected in January.

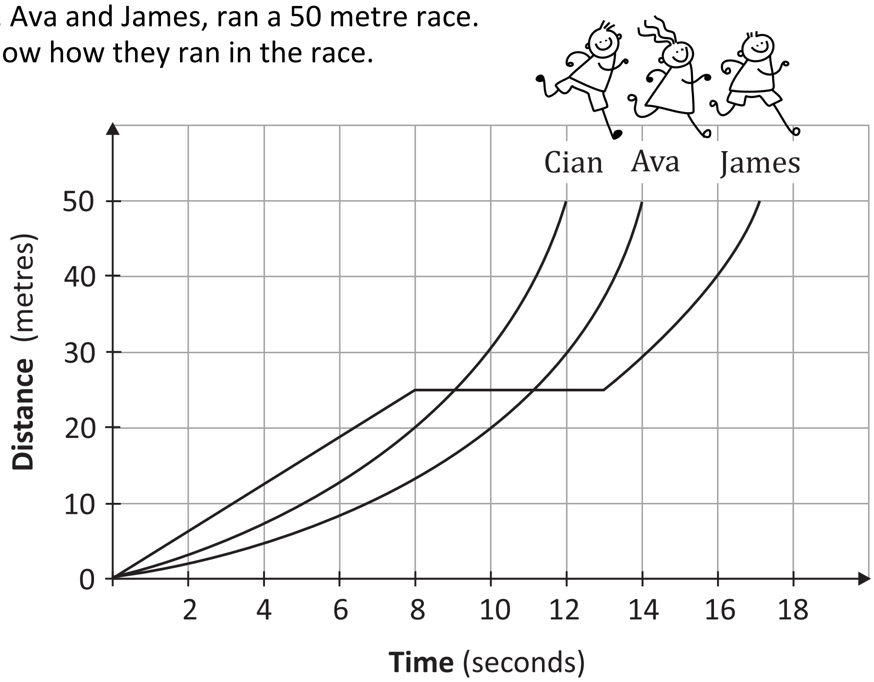
Note: There are 31 days in January.

Least Number of Eggs: Greatest Number of Eggs:

(c) Write $\dfrac{4^7 \times 4^6}{4^2 \times 4^3}$ in the form 4^n, where $n \in \mathbb{N}$.

Question 12 **(Suggested maximum time: 10 minutes)**

Three athletes, Cian, Ava and James, ran a 50 metre race.
The graphs below show how they ran in the race.



Based on the graphs, answer the following questions.

(a) How long did it take James to finish the race? Answer =

(b) Which athlete won the race?
Justify your answer. Answer =

Justification:

(c) Which athlete stopped during the race?
Justify your answer. Answer =

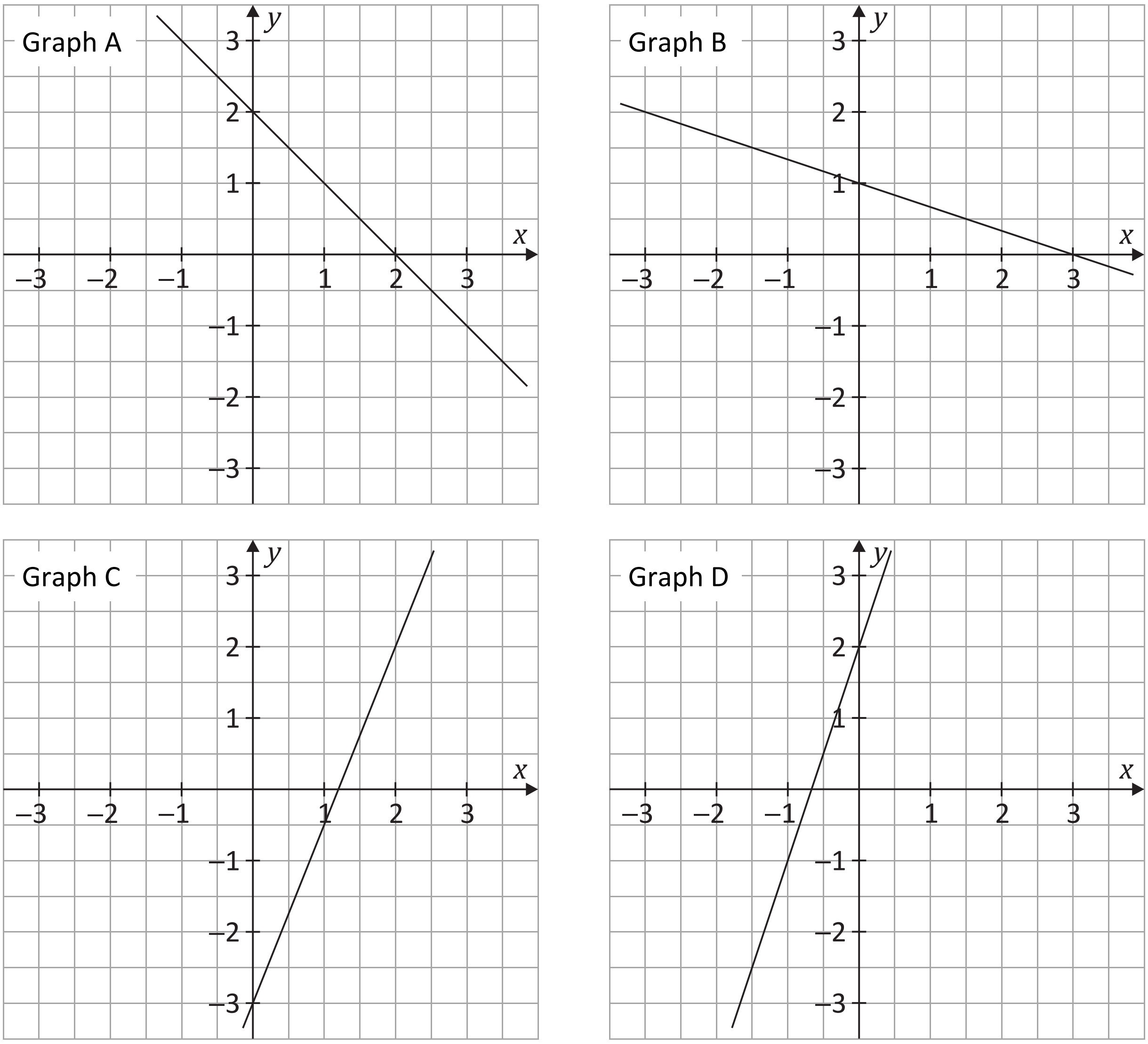
Justification:

(d) How long did this athlete stop for? Answer =

(e) Find Cian's average speed for the race in metres per second, correct to one decimal place.

Question 13 **(Suggested maximum time: 5 minutes)**

The graphs of four functions are shown below.



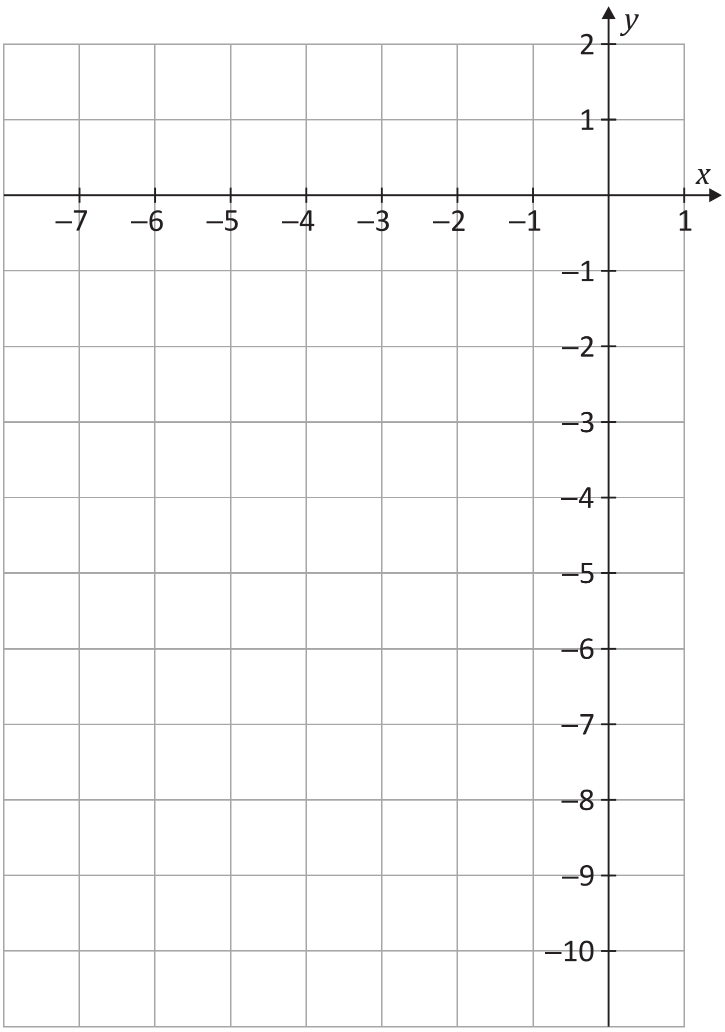
Write the letters ***A***, **B**, **C** and **D** into the table below to match each function with the correct graph. **Justify** each of your answers.

Function	Graph	Justification
$y = 3x + 2$		
$y = -\frac{1}{3}x + 1$		
$y = -x + 2$		
$y = \frac{5}{2}x - 3$		

Question 14 **(Suggested maximum time: 10 minutes)**

(a) Draw the graph of the following function in the domain $-6 \leq x \leq 0$, for $x \in \mathbb{R}$.

$$y = x^2 + 6x$$



(b) $(-6, -10)$, $(-3, -7)$ and $(0, -4)$ are three couples of the function $y = x - 4$.

By plotting the three couples on the same axes used for part **(a)**, draw the graph of the function $y = x - 4$, in the domain $-6 \leq x \leq 0$, for $x \in \mathbb{R}$.

(c) Use the graphs to write down the points of intersection of $y = x^2 + 6x$ and $y = x - 4$.

Answer = $(\quad , \quad)$ and $(\quad , \quad)$

Question 15 **(Suggested maximum time: 10 minutes)**

(a) Lisa works in a local shop from $8{:}00$ am to $12{:}00$ noon and from $2{:}00$ pm to $5{:}00$ pm each day, Monday to Friday. She is paid €$10{\cdot}75$ per hour.

(i) How many hours per week does Lisa work?

(ii) Find Lisa's **gross pay** for the week.

(b) Lisa sold a cake in the shop. It cost €$9{\cdot}08$ **after** VAT at $13{\cdot}5\%$ was added.
Find the cost of the cake **before** VAT was added.

(c) The shop bought the cake from a local bakery for €$5{\cdot}20$.

(i) Work out the **profit** the shop made on the cake.

(ii) Work out the **profit percentage** on the cake (as a percentage of the cost price).
Give your answer correct to one decimal place.

Page for extra work.
Label any extra work clearly with the question number and part.

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